

# Department of Computer Science

**Course:** AI Programming

**Lab Number:** Lab 2: Boolean Expressions and Type Evaluation in Python

## 1. Objective

By the end of this lab, students will be able to:

- Understand and evaluate **Boolean expressions** in Python
- Apply logical operators: and, or, not
- Analyze **short-circuit evaluation** behavior
- Identify **truthy and falsy values**
- Understand how Python handles **non-Boolean operands** in logical expressions
- Perform **type conversion** using int(), float(), and bool()
- Use type() to determine data types

## Task 1

1. Write a python script named “q2.py” to evaluate all of the following expressions. Then fill the following table (submit both your q2.py and the Table below):

| Expression               | Result |
|--------------------------|--------|
| 3 < 5                    |        |
| 3 < 5 and 5 < 3          |        |
| True and False           |        |
| True or False            |        |
| not True                 |        |
| not False and True       |        |
| True and False and True  |        |
| True or (False and True) |        |
| (5/0==1) and False       |        |
| False and (5/0==1)       |        |
| bool(1 and 2)            |        |
| bool(5 and 0)            |        |

## Task 2:

2. Write a python script named “q3.py” to evaluate all of the following expressions, then fill the following table (submit both your q3.py and the Table below):

| Expression                    | Result |
|-------------------------------|--------|
| <code>float(4)</code>         |        |
| <code>int(5.3)</code>         |        |
| <code>int(True)</code>        |        |
| <code>float(int(5.3))</code>  |        |
| <code>float(7) // 4</code>    |        |
| <code>int(7 / 4)</code>       |        |
| <code>6.2 and False</code>    |        |
| <code>True and 6.2</code>     |        |
| <code>type(4.5)</code>        |        |
| <code>type(True and 3)</code> |        |

## Task 3

3. If your answers are correct, then you will notice that there are some weird things going on with the “and” and “or” operators. How does this operator work if one of the two operands is not a Boolean?